

CLASS-8

LESSON-4 DATA HANDLING

(This PDF Based on NCERT Book)

DATA HANDLING(डेटा संधारण)-

1. The Core Purpose: Why Graphs?

If I tell you a list of 50 temperatures for the month, you'll get confused. But if I show you a line that goes up and down, you immediately see when it was hot and when it was cold. Graphs help us see **patterns, trends, and totals** at a single glance.

2. Common Types of Graphs

Different "stories" require different types of graphs. Here are the most common ones:

A. Pictograph (The Picture Graph)

This uses symbols or pictures to represent a specific number of items.

- **Best for:** Very simple data, like counting fruit or animals.
- **Key:** Every pictograph has a "Key" (e.g., 1  = 10 apples).

B. Bar Graph (The Comparison Graph)

This uses rectangular bars to show how large each category is. The taller or longer the bar, the higher the value.

- **Best for:** Comparing different groups (e.g., "Which sport is most popular in our school?").
- **Parts:** It has a **Horizontal Axis** (Categories) and a **Vertical Axis** (Numbers).

C. Pie Chart (The "Whole" Graph)

A circle represents 100% of the data. Each "slice" shows how much of the total a certain part takes up.

- **Best for:** Showing fractions or percentages (e.g., "How do I spend my monthly budget?").

D. Line Graph (The Change Graph)

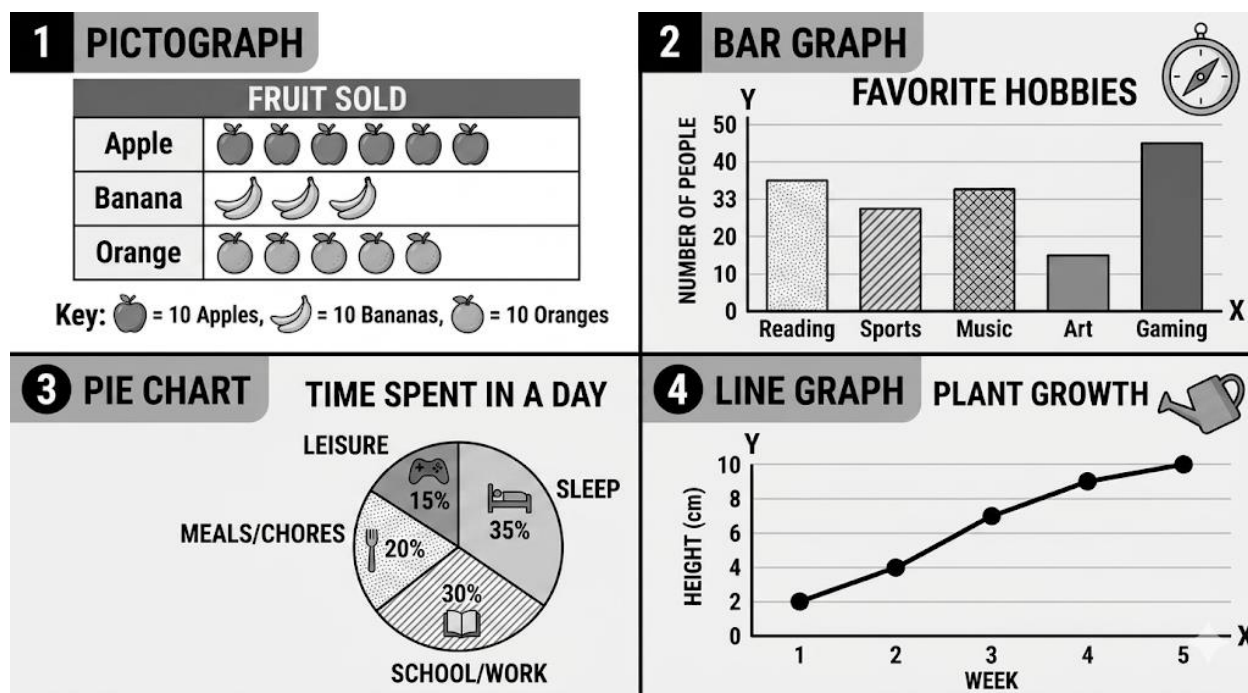
This uses points connected by a line. It is perfect for showing how something changes over **time**.

- **Best for:** Tracking progress (e.g., "How much did my height grow every year?").

3. The 4 Steps of Data Handling

To make any of the graphs above, you must follow this sequence:

1. **Collection:** Gathering your raw info (e.g., counting how many cars pass by your house).
2. **Organization:** Using a **Frequency Table** or **Tally Marks** to group the data so it's not messy.
3. **Representation:** Choosing the right graph (Bar, Pie, or Line) and drawing it.
4. **Interpretation:** Looking at your graph and asking, "What does this tell me?" (e.g., "Traffic is heaviest at 5:00 PM").



DRAW A PIE CHART(पाई चार्ट बनाना)-

Drawing a pie chart is like cutting a pizza into slices based on how much of a "whole" each category takes up. To do this accurately, you have to convert your data into **degrees** because a full circle is always **360°**.

Here is the step-by-step process using a simple example.

The Example Data: Favorite Snacks

Imagine you ask 20 friends what their favorite snack is:

- **Chips:** 10 people
- **Fruit:** 6 people
- **Chocolate:** 4 people
- **Total:** 20 people

Step 1: Calculate the Fraction

First, find out what part of the total each snack represents.

- **Chips:** $10 / 20 = 0.5$
- **Fruit:** $6 / 20 = 0.3$
- **Chocolate:** $4 / 20 = 0.2$

Step 2: Convert to Degrees

Since a circle has 360° , multiply each fraction by 360 to find the size of the "slice."

- **Chips:** $0.5 \times 360 = 180^\circ$ (Exactly half the circle)
- **Fruit:** $0.3 \times 360 = 108^\circ$
- **Chocolate:** $0.2 \times 360 = 72^\circ$

(Check: $180 + 108 + 72 = 360$. If it adds up, you're ready!)

Step 3: Draw the Chart

1. **Draw a circle** using a compass and mark the center point.
2. **Draw a straight radius** from the center to the top (12 o'clock).
3. **Use a protractor** to measure the first angle (180° for Chips) starting from that radius.
4. From the new line you just drew, measure the next angle (108° for Fruit).
5. The remaining space will automatically be your last category (72° for Chocolate).

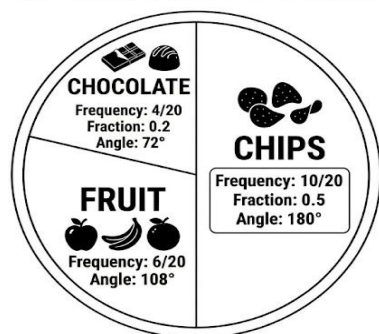
Step 4: Label and Color

Color each slice differently and add a **Label** or a **Legend** so people know what each color means. Adding the percentage (e.g., "Chips 50%") makes it even easier to read.

Summary Table for Drawing

Snack	Frequency	Fraction	Calculation	Angle (Degrees)
Chips	10	$1/2$	0.5×360	180°
Fruit	6	$3/10$	0.3×360	108°
Chocolate	4	$1/5$	0.2×360	72°
Total	20	1		360°

FAVORITE SNACKS: A PIE CHART EXAMPLE



CHANCE AND PROBABILITY(संयोग और संभावना)-

1. What is Chance?

Chance is the informal, everyday word we use to describe the possibility of an event. We often use words like "Impossible," "Unlikely," "Even Chance," "Likely," or "Certain."

- **Example:** If you toss a coin, there is an **even chance** it will land on heads.

Probability Scale



2. What is Probability?

Probability is the mathematical way of measuring chance. Instead of using words, we use numbers (fractions, decimals, or percentages) between **0** and **1**.

- **0** means the event is impossible.
- **1** means the event is 100% certain.

3. The Basic Formula

To find the probability of an event happening—let's call the event **E**—we use this simple formula:

$$P(E) = \frac{\text{Number of favorable outcomes}}{\text{Total number of possible outcomes}}$$

Breaking it down:

- **Favorable outcomes:** The specific result you are looking for.
- **Total outcomes:** Every possible thing that could happen.

4. Simple Examples

A. Rolling a Die

Imagine you want to roll a **4** on a standard 6-sided die.

- **Favorable outcome:** Only one (rolling a 4).
- **Total outcomes:** Six (1, 2, 3, 4, 5, or 6).

$$P(\text{rolling a 4}) = \frac{1}{6}$$

B. Picking a Marble

Imagine a bag with **3 Red** marbles and **7 Blue** marbles. What is the probability of picking a Red one?

- **Favorable outcomes:** 3 (the Red marbles).
- **Total outcomes:** 10 (3 Red + 7 Blue).

$$P(\text{Red}) = \frac{3}{10} = 0.3 = 30\%$$

5. Important Rules to Remember

1. **The Sum Rule:** If you add up the probability of all possible outcomes, the total will always be **1**.
 - Example: In a coin toss, $P(\text{Heads}) + P(\text{Tails}) = 0.5 + 0.5 = 1$.
2. **The Complement Rule:** The probability of something **not** happening is 1 minus the probability of it happening.
 - If the chance of rain is 0.2, the chance of **no rain** is $1 - 0.2 = 0.8$.

EQUALLY LIKELY OUTCOMES(समान रूप में संभावित परिणाम)-

An **Equally Likely Outcome** occurs when every possible result of an experiment or action has the exact same chance of happening. No single outcome is more "favored" than the others.

1. The Core Concept

If an experiment has n possible outcomes, and they are all equally likely, the probability of any one specific outcome occurring is:

$$P(E) = \frac{1}{n}$$

2. Classic Examples

The Coin Toss

When you flip a fair coin, there are 2 possible outcomes: **Heads** or **Tails**.

- Since the coin is balanced, one side isn't heavier than the other.
- $P(\text{Heads}) = \frac{1}{2}$
- $P(\text{Tails}) = \frac{1}{2}$

Both outcomes are **equally likely**.

The Rolling Die

A standard six-sided die has the numbers {1, 2, 3, 4, 5, 6}.

- Assuming the die is not "loaded" (weighted), you have the same chance of rolling a 1 as you do a 6.
- The probability for any single number is exactly $1/6$.

A Balanced Spinner

Imagine a circle divided into 4 equal quarters, colored Red, Blue, Green, and Yellow.

- Because each color takes up the exact same amount of space (90° of the circle), the spinner is just as likely to land on Red as it is on Green.

3. When are outcomes NOT equally likely?

It is helpful to see the difference. Imagine a bag with:

- **9 Red marbles**
- **1 Blue marble**

If you reach in without looking, the outcomes "Red" and "Blue" are **not equally likely**. You are much more likely to pull out a Red marble because there are more of them.

LINKING CHANCES TO PROBABILITY(संभावनाओ को प्रायिकता से जोड़ना)-

Think of **Chance** as a "feeling" and **Probability** as a "number." Linking them just means putting an exact measurement on how you feel about an event.

Here is the simplest way to see the connection:

1. The Word vs. The Number

We use "Chance" words when talking to friends, but we use "Probability" numbers when doing math. They are like two different languages for the same thing.

- **Chance says:** "It is **Impossible** for a dog to speak English."
- **Probability says:** "The probability is **0**."
- **Chance says:** "There is an **Even Chance** of getting Heads in a coin flip."
- **Probability says:** "The probability is $1/2$ (or **0.5**)."
- **Chance says:** "It is **Certain** that tomorrow will come."
- **Probability says:** "The probability is **1**."

2. How to "Make the Link" (The Counting Rule)

To turn a chance into a probability number, you just count the "wins" versus the "total."

Example: A Bag with 4 Marbles (3 Blue, 1 Red)

- **The Chance (Feeling):** Since most marbles are blue, you have a "Good Chance" of picking Blue. It is **Likely**.
- **The Probability (Link):** You count them.
 - Total marbles = 4
 - Blue marbles = 3
 - The Link = $\frac{3}{4}$ (or **0.75**)

Since **0.75** is high on the number scale, it proves your feeling that picking a blue marble is "Likely."

OUTCOMES AS EVENTS(परिणाम घटनाओं के रूप में)-

Think of an **Outcome** as a single result, and an **Event** as a "collection" of results you are looking for.

In simple words, an **Outcome** is what actually happens, and an **Event** is the goal or the category you are checking.

1. The Single Result (Outcome)

If you roll a die, there are 6 possible **Outcomes**. Each one is a single, simple result:

- You get a 1.
- You get a 2.
- You get a 3.
- You get a 4.
- You get a 5.
- You get a 6.

2. The Group of Results (Event)

An **Event** is often more than one outcome put together because you are looking for a specific "type" of result.

- **Example 1:** You want to roll an **Even Number**.
 - The outcomes that fit this event are **2, 4, and 6**.
 - Here, the Event is "Even Numbers," and it contains 3 outcomes.
- **Example 2:** You want to roll a number **greater than 4**.
 - The outcomes that fit this event are **5 and 6**.
 - Here, the Event is "High Numbers," and it contains 2 outcomes.

3. How to tell them apart

You can remember it like this:

- **Outcome:** One specific thing that happened. (The "What")
- **Event:** What you were hoping for or the rule you set. (The "Goal")

Action	Outcomes (All possibilities)	Event (What you want)	Outcomes in that Event
Tossing a Coin	Heads, Tails	Getting Tails	Tails (1 outcome)
Rolling a Die	1, 2, 3, 4, 5, 6	Rolling a "Prime Number"	2, 3, 5 (3 outcomes)
Picking a Card	All 52 cards	Picking a "Red Card"	All Hearts and Diamonds (26 outcomes)

CHANCE AND PROBABILITY RELATED TO REAL LIFE(वास्तविक जीवन से संबंधित संयोग और संभावना)-

In the real world, we don't always have a math book in our hands, but we use **Chance** and **Probability** every single day to make choices. It's basically our way of guessing how "safe" or "risky" a decision is.

Here is how it works in your daily life:

1. Weather Forecasts

When you check your phone and see a "**30% chance of rain**," that is a probability.

- **The Math:** Out of 100 days with similar weather conditions, it rained on 30 of them in the past.
- **The Choice:** You decide if that "chance" is high enough to carry an umbrella.

2. Sports and Gaming

If you are playing a game like Ludo or Carrom, every move has a probability.

- **Ludo:** You need a **6** to start. The chance is **1 out of 6**.
- **Cricket:** Before the match starts, the **Toss** has an **even chance** (50%) for both teams. It is a fair way to start because the outcomes are equally likely.

3. Shopping and Offers

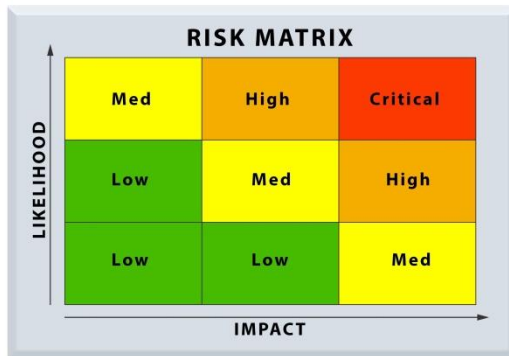
Have you ever seen a "Lucky Draw" or a "Spin the Wheel" offer in an app?

- If there are 1,000 people entered and only 1 prize, your probability is **1/1000** (Very Unlikely).
- Apps use these numbers to keep you interested, even if the "chance" of winning the big prize is very small.

4. Safety and Risk

Think about wearing a helmet or a seatbelt.

- We know that the **probability** of getting into an accident is low on any single day.
- However, we wear a helmet because if an accident *does* happen, the **chance** of a serious injury is much lower with protection. We are managing the "risk" based on probability.



Risk Matrix concept with impact and likelihood - 3d rendering

5. Medical Tests and Health

Doctors use probability to help you get better.

- If a medicine works for **90%** of people, that is a high probability.
- When you take a test, the "accuracy rate" (e.g., 99% accurate) tells you the probability that the result is correct.